Einstein realized that something similar holds for spacetime. Even though two observers in relative motion slice up spacetime in different ways, there are things they still agree on. As a prime example, consider a straight line not just through space, but through spacetime. Although the inclusion of time makes such a trajectory less familiar, a moment's thought reveals its meaning. For an object's trajectory through spacetime to be straight, the object must not only move in a straight line through space, but its motion must also be uniform through time; that is, both its speed and direction must be unchanging and hence it must be moving with constant velocity. Now, even though different observers slice up the spacetime loaf at different angles and thus will not agree on how much time has elapsed or how much distance is covered between various points on a trajectory, such observers will, like Marge and Lisa, still agree on whether a trajectory through spacetime is a straight line. Just as the geometrical shape of the painted trail to the Kwik-E-Mart is independent of the street/avenue slicing one uses, so the geometrical shapes of trajectories in spacetime are independent of the time slicing one uses.[10]

This is a simple yet critical realization, because with it special relativity provided an absolute criterion—one that all observers, regardless of their constant relative velocities, would agree on—for deciding whether or not something is accelerating. If the trajectory an object follows through spacetime is a straight line, like that of the gently resting astronaut (a) in Figure 3.7, it is not accelerating. If the trajectory an object follows has any other shape but a straight line through spacetime, it *is* accelerating. For example, should the astronaut fire up her jetpack and fly around in a circle over and over again, like astronaut (b) in Figure 3.7, or should she zip out toward deep space at ever increasing speed, like astronaut (c), her trajectory through spacetime will be curved—the telltale sign of acceleration. And so, with these developments we learn that *geometricalshapes of trajectories in spacetime provide the absolute standard that determines whether something is accelerating.* Spacetime, not space alone, provides the benchmark.